

Sarah Ali

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🌐 [Sarah Ali](#)

WHO AM I?

Electrical Engineer with strong hands-on experience in **Machine Learning and Computer Vision**, specializing in data-driven modeling, classification, and predictive analysis. **Proficient in Python-based ML frameworks, data preprocessing, model training, and performance evaluation**, with practical exposure to computer vision techniques for **real-world applications**. Combines a solid engineering foundation with analytical problem-solving to develop scalable, efficient, and intelligent systems.

RELEVANT PROJECTS

- Project AViA: Aerial Vision in Action**
An embedded perception system for multi-class object detection on UAVs, developed from first principles with a focus on real-time performance and deployment on resource-constrained hardware. The project integrates computer vision algorithms with embedded systems to enable efficient and reliable aerial perception.
- Natural Language Processing with Disaster Tweets**
Natural Language Processing (NLP) model capable of distinguishing between legitimate disaster reports and metaphorical or irrelevant usage of disaster-related terminology on social media
- Number Plate Detection**
Developed a MATLAB-based system to detect vehicle number plates using image preprocessing, edge detection, and morphological operations. Ensured accurate and consistent detection across varying conditions.

EXPERIENCE

- Intern – DroNext SEECS**

(June – September 2025)

 - Designed 3 UAVs in Fusion 360 and 3D printed prototypes.
 - Gained hands-on experience in **flight controllers, ESCs, and 3D design**.
- Quality Assurance Engineer – 9t5 Pvt. Ltd., Lahore**

(June – September 2024)

 - Designed and executed test plans and cases for web and mobile apps.
 - Performed **manual and automated testing** (functional, regression, performance).

EDUCATION

Degree	Year	Institute	Grades
BS. Electrical Engineering	2026	NUST, Islamabad	CGPA: 3.3

PROJECTS

Face Recognition (Machine Learning on Matlab)	Project AViA: Aerial Vision in Action (computer vision and machine learning based)
Number Plate Detection	Video Action Recognition with Video Transformers
Wireless Charger on a PCB	Digital Multimeter
PID Tuned Flight Controller of Bicopter	Arduino based Bluetooth Control Car
Voltage Overcharge Protection	Digital Voting Machine on FPGA

TECHNICAL SKILLS

- Programming & ML Frameworks:** Python, NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch
- Computer Vision:** OpenCV, image preprocessing, object detection, segmentation, feature extraction
- Machine Learning & AI:** Supervised & unsupervised learning, classification, regression, model training, evaluation

- **Embedded & Edge Deployment:** Integrating ML/CV models on UAVs and resource-constrained hardware
 - **Data Handling & Analysis:** Data cleaning, augmentation, visualization, statistical analysis
 - **Tools & Simulation:** MATLAB, Simulink, Jupyter Notebook, VS Code, Git version control
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